Project Coursework: M.Sc. Computational Methods in Engineering

Buckling Analysis of Cylindrical Shells Using Bogner-Fox-Schmit (BFS) Elements

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Abstract—Cylindrical shells are central to aerospace and civil structures, yet their tendency to buckle suddenly under axial load makes reliable stability prediction essential to safe design. While several finite-element techniques achieve the slope continuity that thin-shell bending theory demands, the Bogner-Fox-Schmit (BFS) element reaches this continuity with a comparatively compact formulation. Its cylindrical variant, however, has received only limited attention in the literature—its original 1967–68 formulation and a short corrective note five years later—and has never been checked against a modern commercial solver. This project addresses that gap by recasting the element formulation for linear eigenvalue buckling analysis, coding the resulting stiffness matrices in MATLAB, and benchmarking the predicted critical loads for a fixed-base cylinder against an ANSYS shell model over a sequence of refined meshes. The study will assess whether the element's convergence is competitive with that of standard commercial shell elements, providing the first independent modern evaluation of the cylindrical BFS element and laying the groundwork for a subsequent extension to shells with cutouts via cut BFS elements.

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1. Introduction

Thin-walled cylindrical shells, widely used in civil and aerospace structures, are prone to sudden buckling under axial compression with severe loss of load-carrying capacity [1]. Conventional shell elements achieve the required C^1 continuity—continuity of both the deflection and its first derivatives (slopes) across element boundaries, needed because thin-plate and shell bending theory involves second derivatives of the deflection in the curvature and strain-energy expressions [2]—only through complex formulations such as TUBA triangles or isogeometric NURBS discretizations [3]. The Bogner-Fox-Schmit (BFS) element offers a simpler route: for a flat plate, tensor products of cubic Hermite polynomials give exact C^1 continuity with few degrees of freedom [2], [4].

This project formulates, implements, and validates a BFS-based linear eigenvalue buckling analysis of cylindrical shells against a commercial ANSYS model. Extending the element from a flat plate to curved, cylindrical geometry changes its behavior: for a flat plate, bending and membrane action are linearly decoupled, but for a cylinder they are coupled [2], [5]. This coupling requires the membrane displacements to be interpolated at the same order as the transverse deflection, which raises the cylindrical element's degree-of-freedom count beyond the flat-plate case [6]. Despite this increase, the element still achieves exact C^1 continuity—something that would otherwise require TUBA- or NURBS-level complexity—and has been shown to converge monotonically [7]. This retained continuity and convergence, rather than a reduced degree-of-freedom count, is the element's real advantage. It is intended as the basis for future work extending it to shells with cutouts using cut BFS (CutBFS) elements and the finite cell method.

Different end conditions appear in the shell-buckling literature for the axial-compression case: the classical closed-form solution for the critical stress assumes simply supported ends [5], while a validated numerical cylinder-buckling benchmark fixes both rims except for the axial degree of freedom of the loaded rim [8]. In this project, the base is fixed and the loaded top edge is left free (Figure 1). The choice of end condition is known to strongly affect the critical load [5].

2. State of the Art

The BFS element originates with an interelement-compatible stiffness/mass formulation [4], extended to curved geometry [7] and to nonlinear postbuckling of flat-plate and cylindrical-shell panels [6]—still its most advanced application. The only other application found is a corrective note [9], after which the cylindrical BFS element does not appear to have been revisited.

Achieving C^1 continuity is not unique to the BFS element, and the available approaches trade off differently. TUBA-family elements offer a fully conforming, mesh-topology-independent formulation, but at high computational cost, requiring a large number of degrees of freedom per element [3]. Isogeometric (NURBS-based) discretizations obtain higher-order continuity directly from the smoothness of the spline basis and are widely used today, but require additional trimming or fictitious-domain treatment once cutouts or other nonconforming features break the smooth parametrization [3].

The BFS element instead achieves exact C^1 continuity with comparatively few degrees of freedom—in its flat-plate form, four per node: displacement, two slopes, and the mixed second derivative—and converges reliably and efficiently, but this simplicity relies on a strict topological requirement (exactly four elements meeting at each interior node, i.e., a regular mesh), which restricts its direct use to simple geometries [3], [10].

This mesh-regularity restriction is addressed by a cut-BFS formulation [10], extended to plate buckling with cutouts via the finite cell method and validated against Abaqus [3], [11]. A vectorized MATLAB BFS implementation is available [12]; isogeometric cylindrical-shell buckling has been validated against commercial FE [8]; meshing near curvilinear boundaries without cut elements has also been addressed [13].

A long-standing gap between theoretical and experimental buckling loads has been attributed to imperfection sensitivity, documented for both torsion [14] and axial compression [5]. Scatter between nominally identical test specimens persists even today, motivating continued stochastic imperfection modeling [1]. Independent of that open question, however, a more fundamental gap exists at the level of the element formulation itself: the original cylindrical BFS formulation [6] predates these modern techniques, none of which has been applied to the cylindrical BFS element itself—which does not appear to have been revisited since [9], let alone validated against a commercial FE code. This project takes up that opportunity for the linear buckling case: reformulating the cylindrical BFS element's geometric stiffness matrix for a generalized eigenvalue buckling problem, implementing it in MATLAB, and validating the resulting critical loads against a

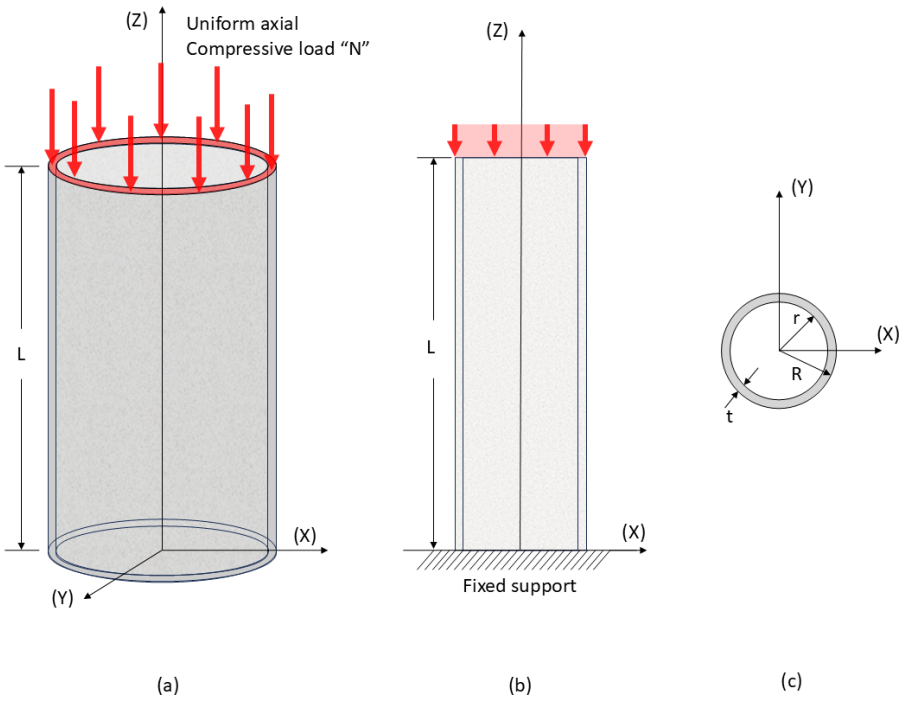


Figure 1. Thin-walled isotropic cylindrical shell under uniform axial compression, shown in the global Cartesian frame (X, Y, Z). (a) 3-D view: the uniform axial compressive load N is applied around the top edge. (b) Elevation view: the base is a fixed support (all translations and rotations restrained); the loaded top edge is free. L is the shell length. (c) Wall cross-section showing the outer radius R, inner radius r, and wall thickness $t = R - r$ ($t \ll R$).

commercial ANSYS model—a modern verification the element has not previously received.

3. Open Research Question

Can a modern, verifiable implementation of the Bogner-Fox-Schmit cylindrical shell element reproduce commercial-FE buckling predictions for axially loaded cylindrical shells, with accuracy and convergence competitive with standard shell elements? Neither the original formulation [6] nor the only other known engagement with the element, a brief corrective note [9], addressed this question, and no BFS-based cylindrical shell buckling code has been benchmarked against a current commercial solver.

This gap likely persists because subsequent research moved toward isogeometric and cut-element formulations that sidestep the BFS element's mesh-regularity restriction [3], [10]. Indeed, BFS-shell compatibility is explicitly named as an open extension of related cut-BFS plate work—one this project takes up for the cylindrical case [11]. Closing this gap requires three steps: deriving a geometric stiffness matrix for the eigenvalue formulation, since the element's 1968 formulation solved the nonlinear postbuckling problem directly by energy minimization rather than as a linear eigenvalue problem [6]; mapping boundary conditions onto its degrees of freedom (the displacement value, two first derivatives, and mixed second derivative of each displacement component), extending the clamped-edge case already given in 1968 [6] to the linear eigenvalue strain-energy formulation [3]; and validating the result against ANSYS, with matched mesh density and boundary conditions to isolate formulation effects.

4. Methodology

4.1. Objectives and Techniques

1. **Formulate** the BFS shell element for cylindrical geometry, following the cylindrical-shell BFS element formulation [6], [7], adapted via the linear eigenvalue strain-energy formulation [3].
2. **Implement** the shape functions and quadrature in MATLAB, using a vectorized approach [12].
3. **Assemble** the elastic stiffness matrix K and the geometric stiffness matrix K_G, and solve the generalized eigenvalue problem

$$(K + \lambda K_G)\, q = 0 \tag{1}$$

for the critical load factor λ and its associated eigenvector q, the vector of nodal degrees of freedom. This discretizes the classical linear plate-buckling theory [15] into finite-element matrix form [3], [16], [17].
4. **Validate** critical loads for an isotropic cylindrical shell with a fixed base and a free loaded top edge against ANSYS, checking mesh-refinement convergence (i.e., confirming the critical load factor approaches a stable value as the mesh is successively refined). Agreement will be measured by the relative error of the critical load with respect to the ANSYS result and by the convergence of the critical load factor as the mesh is refined.

4.2. Simulation Tools

- **MATLAB**: for custom implementation of the BFS element formulation, which no commercial FE package provides natively, following established implementations [3], [12].
- **ANSYS**: the commercial-FE benchmark, using standard shell elements (e.g., SHELL281) for an independent cross-check of the critical load.

4.3. Anticipated Limitations

- The BFS element's regular-mesh requirement restricts the model to simple boundary conditions at this stage (no cutouts, no irregular geometry), consistent with the project's scope.
- Agreement between the MATLAB and ANSYS models alone cannot fully rule out shared modeling assumptions.

4.4. Resource Requirements

- MATLAB with standard linear algebra and eigenvalue-solver toolboxes.
- An ANSYS license with shell-element modeling capability.

5. Outcome

The project will deliver a working MATLAB implementation of the BFS cylindrical shell element for linear buckling analysis, validated against an ANSYS shell model across a mesh-refinement study. The study will establish whether the element converges at a rate competitive with standard shell elements; optimal convergence rates have been reported for BFS-based plate elements [3], [10].

This would re-establish the cylindrical BFS element—unused since its original formulation [6] and the sole corrective note that followed [9]—as a validated modern tool, opening a direct path to the planned CutBFS extension for shells with cutouts [3].

6. Conclusion

This project aims to formulate, implement, and validate a BFS-based linear buckling analysis of cylindrical shells. Its intended contribution is the first modern benchmark of the cylindrical BFS element against a commercial solver: reformulating the element's geometric stiffness matrix for a generalized eigenvalue buckling problem, implementing it in MATLAB, and validating the resulting critical loads against an ANSYS shell model across a mesh-refinement study.

Next steps: (1) derive and code the elastic and geometric stiffness matrices for the cylindrical BFS element [6], [7], adapted to the linear eigenvalue formulation [3]; (2) build the equivalent ANSYS shell model; (3) run the mesh-refinement convergence study and compare critical loads. From there, the next phase extends this validated formulation to shells with cutouts using CutBFS elements.

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